Engineering - Graduate Level Assessment 5 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. If the speed of an airplane is measured by means of a stagnation tube, what correction must be applied to the constant-density indication when the plane is cruising at 300 mph?

   (a) - 10 mph
   (b) 25 mph
   (c) -30 mph
   (d) -15 mph
   (e) - 5 mph

2. The field winding of a shunt motor has a resistance of 110 ohms, and the voltage applied to it is 220 volts. What is the amount of power expended in the field excitation ?

   (a) 660 watts
   (b) 880 watts
   (c) 200 watts
   (d) 120 watts
   (e) 110 watts

3. In an experiment involving the toss of two dice, what is the probability that the sum is 6 or 7?

   (a) 5/36
   (b) 3/12
   (c) 6/36
   (d) 1/6
   (e) 17/36

4. What is the probability of throwing a "six" with a single die?

   (a) 3/6
   (b) 1/12
   (c) 1/6
   (d) 2/6
   (e) 1/8

5. Find the magnitude of E for a plane wave in free space, if the magnitude of H for this wave is 5 amp/meter.

(a) 1200 volts/meter

(b) 5 ohms

(c) 1883 volts/meter

(d) 940.5 volts/meter

(e) 5 volts/meter

True / False (10 marks)

Answer True or False

6. True or False: The impedance of a lossless medium with _r = 10 and _r = 5 is 266 ohms.

7. True or False: A combinational PLD with a fixed AND array and a programmable OR array is referred to as a Field Programmable Gate Array (FPGA).

8. True or False: The lowest critical frequency of a network occurs at zero for an RL circuit, indicating its presence at or near the origin.

9. True or False: The electric field produced by a varying magnetic field B = B_0e^bt is given by E = bB_0e^btra_phi.

10. True or False: The partial-fraction expansion of F(s) = (8s + 2) / (s^2 + 3s + 2) is F(s) = [(-2) / (s + 1)] + [(-8) / (s + 2)].

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the quantization process of a signal and derive the mean-square quantization error, considering the error uniformly distributed in the range -S/2 to +S/2.

12. Given a sphere of radius a with a uniform charge density inside and zero outside, derive the electric potential throughout the entire space by solving Poisson's equation. Discuss the implications of your solution.

End of Paper